$y_1 = -10$ and $y_2 = -2$.

The required mid-point is $\left(\frac{-8+4}{2}, \frac{-10-2}{2}\right)$ or $(-2, -6)$.

Let us now see the application of mid-point formula in our real life situation, consider the longitude and latitude of the following cities.

Name of the city	Longitude	Latitude
Chennai (Besant Nagar)	80.27° E	13.00° N
Mangaluru (Kuthethoor)	74.85° E	13.00° N
Bengaluru (Rajaji Nagar)	77.56° E	13.00° N

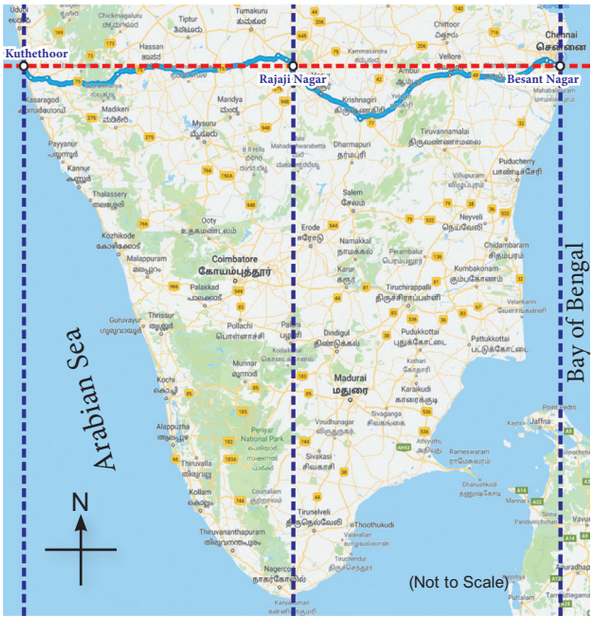


Fig. 5.23

Let us take the longitude and latitude of Chennai (80.27° E, 13.00° N) and Mangaluru (74.85° E , 13.00° N) as pairs. Since Bengaluru is located in the middle of Chennai and Mangaluru, we have to find the average of the coordinates, that is $\left(\frac{80.27+74.85}{2}, \frac{13.00+13.00}{2}\right)$. This gives (77.56°E, 13.00°N) which is the longitude and latitude of Bengaluru. In all the above examples, the point exactly in the middle is the mid-point and that point divides the other two points in the same ratio.

Example 5.12 The point $(3, -4)$ is the centre of a circle. If AB is a diameter of the circle and B is $(5, -6)$, find the coordinates of A.

Solution Let the coordinates of A be (x_1, y_1) and the given point is $B(5, -6)$. Since the centre is the mid-point of the diameter AB, we have

$$\frac{x_1 + x_2}{2} = 3 \qquad \frac{y_1 + y_2}{2} = -4$$
$$x_1 + 5 = 6 \qquad y_1 - 6 = -8$$
$$x_1 = 6 - 5 \qquad y_1 = -8 + 6$$
$$x_1 = 1 \qquad y_1 = -2$$

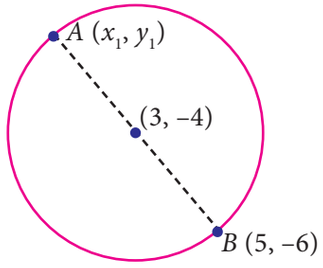


Fig. 5.24

Therefore, the coordinates of A is $(1, -2)$.

5-Coordinat_1.indd 205 02/12/2022 21:45:06